Three obstructions to Stein variational gradient descent
on a MAGI posterior

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Abstract

Stein variational gradient descent (SVGD) is an attractive candidate for MAGI [1] posteriors: it is deterministic, it parallelises trivially, and it needs only the score. We spent four investigations on it and recommend against it. This report sets out why, because the reasons are more interesting than the verdict and two of them are not specific to MAGI.

There are three obstructions, and each would be sufficient alone. First, the variance collapse is dimensional. On these problems d runs from 106 to 608, and the equilibrium variance ratio under the median-heuristic bandwidth is $\ln K/d$ — measured to three decimals over $d \in [50, 608]$ and $K \in [10, 3200]$ — so correct dispersion needs $K \sim e^d$. Ba et al. [3] prove the $\Theta(1/d)$ scaling; the $\ln K$ numerator is an artifact of the $1/\log K$ factor that implementations following Liu and Wang [2] apply, and dropping it is worth a factor of $0.582K/\ln K$, or 39 at $K = 400$. Second, that obstruction *is* removable: at a large fixed bandwidth in coordinates whitened by the Laplace metric, SVGD has a genuine attractor, the residual error is a pure scale deficit, and the Stein identity measures that deficit without a reference — rescaling by $1/\sqrt{R}$ brings the ensemble to 0.57 to 0.96 times the reference chain's own agreement floor. Third, and fatally, the fixed point so obtained is displaced in the *parameters*, along the axis from the joint mode to the posterior mean, by that axis's own length. It is exact for a Gaussian target and biased for anything else. That displacement is precisely the correction the deterministic pipeline exists to compute, so SVGD returns the quantity being estimated.

We also report a bug of our own that invalidated an earlier verdict on preconditioning, two of our own hypotheses that measurement overturned, and the three things from this work worth keeping: the $1/\log K$ deletion, the Stein dispersion ratio as a diagnostic, and the design rule $K \gtrsim e^p$ for SVGD on a p-dimensional marginal.

1 What would count as success

MAGI places a Gaussian process prior on each state component of $\dot{x} = f(x, \theta)$ and constrains it to satisfy the dynamics on a discretisation grid. The unknown is $x = (\theta, X) \in \mathbb{R}^d$ with $d = p + nD$; across the four systems we use, d runs from 106 to 608 and p from 3 to 7. Almost all of the dimension is state.

Three of the four systems have a usable reference: 32 preconditioned NUTS chains, 96,000 draws, $\widehat{R} \le 1.007$. Hes1 does not — $\widehat{R} = 1.76$ with 13% divergences — and is excluded from every quantitative statement here, because without a reference nothing measured on it can be scored.

Scoring is by energy distance in Mahalanobis coordinates, always reported against its own **floor**: the same statistic between two independent subsamples of the reference at the ensemble's particle count. A value of $1.0\times$ the floor means the ensemble is as close to the reference as

$d\backslash K$	50	100	400	1600
50	3.915	4.512	5.452	8.602
100	3.912	4.605	5.774	7.924
200	3.912	4.582	5.914	7.756
325	3.917	4.601	5.956	8.072
608	3.890	4.605	5.978	7.469
$\ln K$	**3.912**	**4.605**	**5.991**	**7.378**

Table 1: $d \cdot \mathrm{Var}_{\mathrm{SVGD}} / \mathrm{Var}_{\mathrm{target}}$ against $\ln K$, standard RBF kernel under the median heuristic with the $1/\log K$ factor. Prodigy, 2000 iterations, started at exact draws.

the reference is to itself, which is the most a finite reference can establish. Marginal standard deviations are *not* used as a score, and this is the specific trap the problem sets: an early run held every marginal sd at 0.991 of the reference — visually perfect credible intervals — while sitting 45× the energy floor from the target. SVGD's collapse is anisotropic and no marginal sees it.

The incumbent is the deterministic pipeline of [6]: profile the states out by Laplace at each θ, integrate the p-dimensional remainder by importance sampling. It reaches the reference's own agreement floor on all three scorable systems — largest parameter error 0.0126 against a floor of 0.0100, 0.0093 against 0.0081, 0.0119 against 0.0405 — in four to six seconds. Any case for SVGD has to be made against that.

2 Obstruction 1: the collapse is dimensional

2.1 The original observation, and the control that explains it

Started *at* a correct posterior sample on FitzHugh–Nagumo and run, SVGD moves away from it: the energy distance degrades from 0.085 to 6.89 under the standard RBF kernel and 3.80 under the density-reweighted kernel of [4], and the Stein-identity ratio falls from 1 to 0.019 and 0.24. Since the starting point was correct, the motion is unambiguously away from it.

We re-ran that after correcting a Gaussian-process hyperparameter fit and changing the benchmark integrator from forward Euler to RK4, expecting the numbers to move. They do not: the standard kernel reproduces to three significant figures (energy 6.895 → 6.847, Stein 0.0190 → 0.0194), and the explicit 2×2 of {RK4, Euler} × {fixed, old} hyperparameters gives Stein 0.0192 to 0.0194 in every cell. Neither correction has any bearing on it.

The reason they have no bearing is that the effect is not a property of the MAGI posterior at all. The control that settles it is embarrassingly simple and had not been run: SVGD on $\mathcal{N}(0, I_{325})$, started at exact draws. It collapses to a flat 1.8% of the correct variance — 89× the floor, the same as on the real posterior. No MAGI, no anisotropy, no non-Gaussianity.

2.2 The law

Sweeping d and K on isotropic Gaussian targets, started at exact draws so there is no burn-in to mistake for an equilibrium, the mean variance ratio times d collapses onto a function of K alone:

$$\frac{\mathrm{Var}_{\mathrm{SVGD}}}{\mathrm{Var}_{\mathrm{target}}} = \frac{\ln K}{d}, \tag{1}$$

	bandwidth	law	at $K = 400$, $d = 800$
Ba et al. [3], Cor. 4	$h = \text{Med}$	$0.582\,K/d$	0.291
this work, and what implementations ship	$h = \text{Med}/\ln K$	$\ln K/d$	0.0075

to three decimals over an order of magnitude in d, confirmed separately by sweeping K from 10 to 3200 at fixed d. At $d = 325$ and $K = 400$ that is 1.8%.

The mechanism is a feedback loop in the bandwidth rule. The median heuristic measures h from the *ensemble*, so contraction tightens the bandwidth, which weakens the repulsion at the scale that matters, which permits further contraction. The equilibrium is where that loop closes, and it closes at a normalised bandwidth of $h \approx 2$ regardless of where it started.

2.3 What is ours and what is not

The $1/d$ scaling is not ours. Ba, Erdogdu, Ghassemi, Sun, Suzuki, Wu and Zhang [3] prove, for an isotropic Gaussian target in the proportional limit $d/K \to \gamma > 1$ under the *plain* median heuristic $h = \text{Med}$, that $\text{Var}_{\text{SVGD}} / \text{Var} = (e-1)^{-1}K/d = 0.582\,K/d$ (their Corollary 4), conditional on a near-orthogonality assumption they verify numerically rather than prove. Section 2's (1) is the same $1/d$ with a different numerator, and the difference is entirely the bandwidth rule:

Running both conventions through one harness reproduces both laws to within 1% — measured 0.29026 against a predicted 0.29099, and 0.00750 against 0.00749 — which validates the harness against published theory and isolates a fact worth acting on. Liu and Wang [2] recommend $h = \text{Med}/\log K$, and that is what our library used. The cost is $0.582K/\ln K$: a factor of 39 at $K = 400$ and 281 at $K = 4000$, always against the log.

Deleting it is one line. On the three real posteriors at $K = 400$ it improves the energy distance by 1.8 to 2.4× and the Stein ratio by 5 to 9×, better in all six cells tested; on $\mathcal{N}(0, I_{20})$ at $K = 200$ the variance ratio goes from 0.265 to 0.929 against a target of 1. It is free and it is a strict improvement, and we have made the change. It is not a fix: both variants are $O(1/d)$, and (1) rearranged says correct dispersion needs $K \sim e^d$. At $d = 325$ that ends the discussion.

Remark 1. The gain is smaller on real posteriors than on Gaussians (2×, not 39×) because neither law applies exactly to an anisotropic target. The direction is the same in every case measured.

3 Obstruction 2: fixing the bandwidth buys time, and then a scale error

If the adaptive bandwidth is the problem, fix it. A large constant h removes the feedback loop by construction, and on an isotropic Gaussian a fixed $h \approx 100d$ — two orders beyond the $\sigma = \sqrt{d}$ that [3] tests and rejects — has the correct fixed point $v = 1$.

On the real posteriors this works, and then does not. A fixed h converts the collapse from an attractor into a transient: the trajectory is a function of $h\times$ iterations, so ten times the bandwidth buys ten times the delay and nothing else. Two starts a factor of 16 apart in variance meet at the same equilibrium and hold it to three digits for 450,000 iterations — so there *is* a fixed point, it is simply the wrong one.

Remark 2 (a mechanism of ours that was wrong). We first attributed the wrongness of that fixed point to anisotropy, on the grounds that a scalar bandwidth cannot serve directions whose scales differ. That diagnosis survives as a description of where the error goes, but the claim we built on it — that there is no fixed point at fixed bandwidth — was an artifact of reading a single stiff-direction statistic. It is retracted.

3.1 The Laplace metric whitens these posteriors nearly perfectly

If anisotropy is what is left, remove it. Whitening by L with $LL^\top = H^{-1}$ at the mode works better than we expected:

system	$\mathrm{cond}(\Sigma_{\mathrm{ref}})$	$\mathrm{cond}(L^{-1}\Sigma_{\mathrm{ref}}L^{-\top})$	5–95% eigenvalues after
fn	7.9×10^3	**47**	0.86 – 1.22
hiv	$\mathbf{5.6 \times 10^{10}}$	**1.6**	0.82 – 1.21
lorenz	1.7×10^5	192	0.82 – 1.53

HIV's condition number falls by ten orders of magnitude. So preconditioning plus a fixed bandwidth is the configuration to try, and it had never been run: earlier work tested preconditioning only at the adaptive bandwidth, where it was the worst of four kernels.

Remark 3 (a bug of ours). That earlier verdict was worthless, and not because of the bandwidth. Our driver computed the preconditioned score as `L.T @ jax.grad(lambda z: logp(x0 + L @ z))(y)`. Automatic differentiation already differentiates through the `L @ z` inside, so it returns $L^\top \nabla_x \log p$ on its own and the explicit `L.T` applied the metric a second time, giving $L^\top L^\top \nabla_x \log p$. Verified against the analytic chain rule, the corrected line agrees to 9.7×10^{-16} and the old line is exactly the doubly-applied form to 1.2×10^{-15}: a 98% relative error. Every preconditioned measurement we had made was invalid. None had been published.

3.2 With the bug fixed: an attractor, and a pure scale error

Whitened, at $h = 10h^\star$ with $h^\star = 2d/\ln K$, and with a small *fixed* step — Prodigy is unstable in whitened coordinates — SVGD has a genuine attractor. The variance ratio in the stiffest 10% of directions, on FitzHugh–Nagumo at $K = 400$ over 100,000 iterations:

start	5k	25k	50k	100k
correct	0.959	0.787	0.738	**0.748**
narrow 4×	0.077	0.137	0.244	**0.494** ↑
wide 4×	13.18	4.703	1.587	**0.788** ↓

Approached from a factor of four below and a factor of four above. And the residual deficit is now *uniform*: 0.748 against the 0.719 the isotropic law predicts, a 1.04× mismatch, where the unpreconditioned dynamics leaves a 20× mismatch between the mean and the stiff directions. Preconditioning converts a shape error into a scale error.

A scale error is one number, and the Stein identity measures it without a reference. For a Gaussian target and an ensemble of covariance A, $R = \mathrm{tr}(A\Sigma^{-1})/d$, so an ensemble uniformly a factor c too narrow reads $R = c$; rescaling about its own mean by $1/\sqrt{R}$ should correct it, with no reference, no tuning and no extra gradient evaluations. It does:

On the *ensemble*, this is a success and we report it as one. On HIV it beats 400 exact draws on a Kolmogorov–Smirnov comparison. Obstruction 1 is removable.

system	ensemble	Stein R	energy × floor	stiff-var	max $\lvert\theta$ err$\rvert$
fn	`fit()` draws	1.274	1.65	1.058	0.077
	SVGD equilibrium	0.702	2.42	0.748	0.242
	rescaled by $1/\sqrt{R}$	1.074	**0.68**	**1.065**	0.242
lorenz	`fit()` draws	1.019	1.24	1.037	0.071
	SVGD equilibrium	0.653	2.92	0.696	0.401
	rescaled by $1/\sqrt{R}$	1.074	**0.96**	**1.066**	0.401
hiv	`fit()` draws	1.000	1.03	1.074	0.096
	SVGD equilibrium	0.538	6.78	0.573	0.067
	rescaled by $1/\sqrt{R}$	1.000	**0.57**	**1.065**	**0.067**

Table 2: The reference-free recipe: start from `fit()` draws, whiten by H^{-1}, run at fixed h with a small fixed step to equilibrium, measure R, rescale by $1/\sqrt{R}$. A 2.4 to 6.8× ensemble becomes a 0.57 to 0.96× one, from the ensemble alone. Note the last column, which is the subject of Section 4.

4 Obstruction 3: the fixed point is displaced in the parameters

The last column of Table 2 is the reason none of this is a recommendation. The largest parameter error goes from 0.077 to 0.242 on FitzHugh–Nagumo and 0.071 to 0.401 on Lorenz. The rescaling does not touch it, and cannot: it corrects a scale about the ensemble's own mean, and this is an error *in* that mean. Only HIV improves.

The energy distance improves while the parameters degrade because the energy distance is dominated by the 322 trajectory states. MAGI exists to estimate the 3 to 7 parameters.

4.1 Where the displacement points

It is not arbitrary. Decomposing the equilibrium's θ mean along the axis from the joint MAP to the reference mean, with t the signed fraction along that axis and *gap* its length:

system	$t(\theta)$	gap (sd)	$\lvert t\rvert\cdot$gap	measured θ error
fn	+0.186	1.029	0.191	0.242
lorenz	−0.221	1.803	0.399	**0.401**
hiv	+0.454	0.148	0.067	**0.067**

The displacement is aligned with that axis to 0.92–0.999, against 0.58 for a random direction at $p = 3$, and two of the three products reproduce the measured error to the printed digit. **HIV improves because its gap is 0.15 posterior standard deviations, not because it differs in kind.** The effect is specific to θ: over the same run FitzHugh–Nagumo's state block converges *onto* the reference mean (t: $0.427 \to 0.024$) while θ leaves it.

4.2 The mechanism is non-Gaussianity

Run the identical dynamics — same metric, bandwidth, step, start, particle count — on an exact Gaussian with the reference's own mean and covariance. The θ error lands at 0.0045 sd on FitzHugh–Nagumo and 0.0351 on Lorenz, *below* each reference's own Monte Carlo floor, against 0.2422 and 0.4013 on the real posteriors.

So SVGD's fixed point is exact for a Gaussian target, and on a target that is not one it sits displaced along the mode-to-mean axis by a fraction of that axis's length. That axis is not an incidental feature of these problems. Correcting the mode to the mean is what the deterministic pipeline is *for* [5]: the Laplace approximation at the MAP is mis-centred by a full posterior standard deviation on a benchmark as benign as FitzHugh–Nagumo, and removing that bias is the entire content of the method SVGD would be replacing. SVGD gives it back.

Remark 4 (a second hypothesis of ours that was wrong). The natural explanation is mode-seeking: the drift term is a kernel-weighted average of the score, which should pull the ensemble mean toward high density. That predicts $t > 0$ everywhere. It is wrong — Lorenz's t is -0.221, drifting *away* from the MAP and past the reference mean on the far side — and the fraction is not universal either ($+0.186$, -0.221, $+0.454$). The axis and the magnitude are understood; the sign is not.

4.3 What repairs it, and why that settles the question

Importance-reweighting the equilibrium by the profiled log-density recovers parity with the deterministic pipeline: $0.242 \to 0.071$, $0.401 \to 0.090$, $0.067 \to 0.029$, at effective sample sizes of 78%, 66% and 97% and $\widehat{k}$ below 0.3, in negligible time. On HIV that is 3.3× better than `fit()`.

This is the most useful negative result in the report. The repair works by reintroducing $\log \hat{p}(\theta)$ — the profiled density, which is the deterministic pipeline's entire machinery. What does the estimating is the density; SVGD has supplied a proposal. And the incumbent already makes its own proposal, from the profile mode and curvature, at a cost of four to six seconds.

The drift is a finite-K bias and does vanish, at $K^{-0.74}$ on FitzHugh–Nagumo and $K^{-0.36}$ on Lorenz. Reaching `fit()`'s accuracy needs roughly 1.5×10^4 and 1.2×10^5 particles respectively, and `fit()` is more accurate on θ at every K we tested.

Remark 5 (a third hypothesis of ours that was wrong). We had called the step size a real lever and the $\mathrm{lr} \to 0$ limit the most consequential untested gap. At matched flow time — lr×iterations held constant — three step sizes spanning a factor of ten give $t = 0.186$ and a θ error of 0.2422 identically to four digits. The apparent dependence was less flow time. Withdrawn.

5 The one regime where dimension is not the obstacle

Obstruction 1 is about d. The natural response is to run SVGD where d is small: on the profiled p-dimensional marginal, with the states integrated out and autodiff taken through the inner solve. At $p = 3$ this is the one configuration in which SVGD beat the incumbent — 0.30× and 0.36× the 64-draw floor on FitzHugh–Nagumo and Lorenz, better than `fit()` by 2.5 to 8×, at 6 to 12× the wall clock.

It does not survive $p = 5$. On HIV:

K	method	θ energy	× floor	max $\lvert\theta$ err$\rvert$	sd ratio
16	`fit()`	0.1193	**0.50**	0.119	0.961
16	profiled SVGD	0.2529	1.05	0.103	**0.578**
64	`fit()`	0.0514	**1.25**	0.123	0.931
64	profiled SVGD	0.0529	1.29	0.281	**0.881**
256	`fit()`	0.0145	**0.76**	0.103	0.986

The sd ratios say why: the θ ensemble is itself underdispersed, which is (1) reappearing with d replaced by p. It predicts 0.55 at $(K = 16, p = 5)$ and 0.83 at $(K = 64, p = 5)$ against measured 0.33 and 0.78; at $p = 3$ it predicts 0.92 at $K = 16$ and above 1 at $K = 64$, and FitzHugh–Nagumo and Lorenz duly succeed at every K. The law therefore doubles as a design rule, and this is the third thing worth keeping:

$$\text{profiled SVGD needs } \ln K \gtrsim p, \text{ i.e. } K \gtrsim e^p, \tag{2}$$

about 20 particles at $p = 3$ and 150 at $p = 5$, confirmed over seven values of p as $\ln K_{\text{crit}} = 0.86p + 1.45$ with the rate holding to 8–10%. Below it the parameter ensemble collapses exactly as the joint one does.

Two further costs. HIV at $K = 256$ — where the rule says it should work — exhausted a 32 GB V100 in the reverse pass, needing 256 particles × three Cholesky factorisations of a 603×603 block, and had to be rerun with the inner Newton steps rematerialised. And Section 4's displacement applies here too, since the profiled marginal is no more Gaussian than the joint posterior.

6 Verdict

Not worth pursuing, for the joint posterior or the profiled marginal.

The first obstruction alone would not settle it, because it is removable and we removed it. The third does settle it, and it is structural rather than incidental: SVGD's fixed point is exact for a Gaussian and displaced for anything else, along the one axis whose correction is the point of the exercise. No bandwidth, metric or optimiser reaches it, because it is the fixed point and not an approach to it; the only thing that repairs it reintroduces the density that makes SVGD unnecessary. Against an incumbent that reaches the reference's own agreement floor on every scorable system in four to six seconds and reports its own reliability without a reference, there is nothing left to recommend.

Three things are worth keeping, and two of them are not about MAGI:

1. **Drop the** $1/\log K$ from the median-heuristic bandwidth. One line, 1.8 to 2.4× on real posteriors and up to 39× on Gaussians, better in every case measured, and it replaces a rule nobody has analysed with one that has a theorem attached [3].
2. **Report the Stein dispersion ratio.** $R \to 1$ under the target and $R < 1$ when underdispersed; for a Gaussian target it is exactly the mean variance-inflation factor, and in a smoke test it reads 0.9271 where the measured ratio is 0.9270. It is the only cheap diagnostic that sees SVGD's characteristic failure, which the optimiser's own convergence test cannot: the ensemble converges, to the wrong spread.
3. $K \gtrsim e^p$, for anyone running SVGD on a low-dimensional marginal.

7 What would change the verdict, and what we did not test

We would revisit this if the displacement of Section 4 turned out to be removable by something cheaper than the profiled density — but the Gaussian control makes that unlikely, since it shows the displacement is the fixed point's response to non-Gaussianity rather than an artifact of finite K, finite time or the metric. The sign of t is unexplained; explaining it would be satisfying and would not change the magnitude.

Hes1 is untested throughout, and it is the one case where the incumbent is genuinely unreliable: its gate declines, its effective sample size varies from 2.3% to 21.1% with the Sobol seed, and its reference does not converge. It is also the only case where SVGD might have an argument, in that it does not traverse the funnel geometry that defeats NUTS. We did not test it because with no usable reference a success there could not be scored, and because Section 4's mechanism is indifferent to the funnel. The better routes for Hes1 are elsewhere: observe the unobserved component, reparameterise to the profiled coordinates, or Riemannian HMC [6].

Finally, all measurements here are float64, at $K = 400$ unless stated, on two GPUs that agree to four digits on a shared benchmark. Every reference-scored claim excludes Hes1.

References

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